- Monocarpic perennials produce flowers only once during life time but the plants survive for many years. Example: Bamboo.
- Polycarpic perennials produce flowers every year during life time. Example: Coconut.

15.1.1 Kinetics of growth

It is an analysis of the motion of cells or expansion.

1. Stages in Growth rate

The total period from initial to the final stage of growth is called the **grand period of growth.** The total growth is plotted against time and 'S' shaped sigmoid curve (Grand period curve) is obtained. It consists of four phases.

They are:

i. Lag phase
ii. Log phase
iii. Decelerating phase
iv. Maturation phase

i. Lag phase

In this phase new cells are formed from pre-existing cells slowly. It is found in the tip of the stem, root and branches. It is the initial stage of growth. In other words, growth starts from this period.

ii. Log phase or exponential growth

Here, the newly formed cell increases in size rapidly by deposition of cell wall material. Growth rate is maximum and reaches top because of cell division and physiological processes are quite fast. The volume of protoplasm also increases. It results in rapid growth and causes elongation of internode in the stem.

iii. Decelerating phase or Decline phase or slow growth phase

The rate of growth decreases and becomes limited owing to internal and external or both the factors because the metabolic process becomes slow.

iv. Steady state period or maturation phase

In this phase cell wall thickening due to new particle deposition on the inner surface of the cell wall takes place. The overall growth ceases and becomes constant. The growth rate becomes zero.

2. Types of growth rate

The increased growth per unit time is termed as growth rate. An organism or part of an organism can produce more cells through arithmetic growth or geometric growth or both.

i. Arithmetic Growth Rate

If the length of a plant organ is plotted against time, it shows a linear curve and this growth is called **arithmetic growth**.

- The rate of growth is constant and it increases in an arithmetic manner.
- Only one cell is allowed to divide between the two-resulting progeny cell.
- One continues to divide but the other undergoes cell cycle arrest and begins to develop, differentiate and mature.
- After each round of cell division, only a single cell remains capable of division and one new body cell forms.

For example, starting with a single cell after round 1 of cell division there is one dividing cell and one body cell. After round 2 there are two body cells, after round 3 there are three and so on (Figure 15.1).

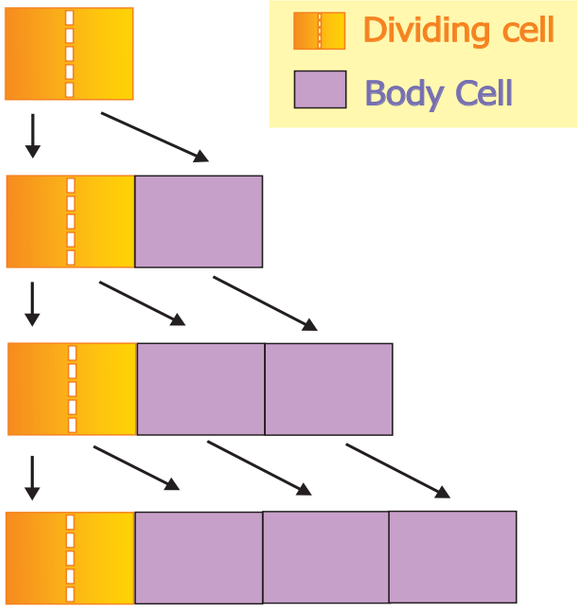


Figure 15.1: Arithmetic Growth Rate

The plants single dividing cell would undergo one million rounds of nuclear and

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